

LESSON 6.11

CONVERTING UNITS USING RATIOS



LESSON INTRODUCTION

MA.6.AR.3.5 Solve mathematical and real-world problems involving ratios, rates, and unit rates, including comparisons, mixtures, ratios of lengths and conversions within the same measurement system.

LEARNING TARGETS

- I can use ratios to convert units within the same measurement system.

BENCHMARK COHERENCE

BUILDING ON

MA.5.M.1.1 Solve multi-step real-world problems that involve converting measurement units to equivalent measurements within a single system of measurement.

WORKING TOWARD

MA.7.AR.3.2 Apply previous understanding of ratios to solve real-world problems involving proportions.

MA.7.AR.3.3 Solve mathematical and real-world problems involving the conversion of units across different measurement systems.

ESSENTIAL QUESTIONS

- How do you use ratios to convert units within the same measurement system?
- How can you represent real-world scenarios using ratio conversions?

LESSON NARRATIVE

In this lesson, students use unit rates to convert various measurements of customary and metric units within those systems. Students explore unit rate conversions for length, volume, mass, and time and apply unit rates to convert measurements. Students analyze errors in another student's work converting units of measure. Then, students practice converting units of measure using rates. Finally, students reflect on their level of confidence using ratios to convert units within the same measurement system.

COMPONENT SUMMARY

6.11.1 WARM-UP (~5 MIN.)

Describe how furniture designers use math

6.11.2 GUIDED INSTRUCTION (20 MIN.)

Use unit rate conversions for length, volume, mass, and time

6.11.3 GUIDED PRACTICE (5 MIN.)

Convert units of measure using unit rates

6.11.4 COLLABORATION (10 MIN.)

Analyze errors in a student's work converting units of measure

6.11.5 YOUR TURN! (5 MIN.)

Practice converting units of measure using unit rates

6.11.6 WRAP-UP (~5 MIN.)

Students reflect on their level of confidence using ratios to convert units within the same measurement system

RESOURCES

GLOSSARY

Key Terms:

- Customary units
- Metric units

Additional Terms:

- Ratio

MATERIALS

- Warm-Up Furniture Designer video
- (Optional) Note cards
- (Optional) Chart paper
- (Optional) Markers

ADDITIONAL PREPARATION

- Warm-Up 6.11.1 contains a video to accompany the question prompts.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may make computational errors when converting units of measurement.
- Students may accidentally use the reciprocal of the conversion factor.

THINGS TO CONSIDER

- Students do not need to memorize conversion rates when engaging in problem-solving. Encourage students to use the tools they have (conversion tables provided as reference charts and in the Student Workbook) in order to solve problems efficiently.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

Consider structuring 6.11.3 Guided Practice and 6.11.5 Your Turn! with the Think Pair Share routine. First, have students solve the problem individually. Then, have students discuss their answers. Tell students to justify their responses and critique each other's reasoning. Discussing the problems will help students synthesize their understanding and build student confidence.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.2.1 Multiple Representations

In 6.11.2 Guided Instruction students use double number lines and equivalent ratio tables to convert units of measurement. Using these visual models will help students organize the quantities, determine the ratios, and accurately complete their calculations. Seeing multiple representations of the equivalent ratios will help students visualize the concepts and consolidate student understanding.

DIFFERENTIATE INSTRUCTION

Consider alternative ways for students to engage with written descriptions or self-reflection like in 6.11.4 Collaboration and 6.11.6 Wrap-Up. Digital tools and apps like Flipgrid offer students differentiated approaches to communicating with others and capturing their personal reflections.

- 6.11.4 Collaboration: Have students verbally record their responses to Valerie using Flipgrid and submit as homework.
- 6.11.6 Wrap-Up: Have students describe what they learned with video, with more creative submissions earning bonus points. (encourage creativity in both examples)

6.11.1 WARM-UP: FURNITURE DESIGNER

Component Overview: Display the “Career Profile – Furniture Designer” video for the entire class. After students watch the video, prompt the students to read and answer the questions in the Warm-Up independently. Discuss how the furniture designer used math in the video with the class.

TEACHER MOVES

Turn and Talk

Engage students in a Turn and Talk, after watching the video, to discuss question 1. Position students to consider the different ways in which Jared used math to create furniture. This will increase investment and engagement with the lesson.



6.11.1 Warm-Up: Furniture Designer

1. Jared Rusten is a woodworker who creates modern furniture in San Francisco.



- a. Jared usually starts designing furniture by sketching his ideas before he makes a scaled drawing using measurements to build the actual piece of furniture. What do you think a scale drawing is?

Answers will vary.

A scaled drawing shows a real object that has accurate sizes that are reduced or enlarged by a scale or specific amount.

- b. Jared said, “I never thought that I’d be using math as much as I do now.” What are some of the ways he uses math as a furniture designer and maker?

Answers will vary.

Scale drawings allow us to visualize larger objects on paper. As a furniture designer and maker, you draft or sketch your designs on paper before creating the pieces. You are also able to scale your designs before they are in actual spaces. He multiplies to enlarge, divides to reduce, and uses area to ensure that the furniture fits in a space.



6.11.2 Guided Instruction: Converting Units of Measure Using Ratio Relationships

Many different units are used to measure lengths, weight, capacity (volume), and time.

- What are some units of measure that are used to take each type of measurement?

Answers will vary, but could include:

Length	Weight	Capacity	Time
Centimeter	Ounce	Liter	Second
Foot	Pound	cm^3	Minute
Mile	Kilogram	Quart	Hour
Inch	Gram	Pint	Day
Meter	Ton	Gallon	Week
Yard		Cup	Year

The United States uses the customary system of measurement. The table shows the unit conversions for units within this system.

- Label each row using the words **Length**, **Weight**, and **Capacity** to indicate what each set of units measures.

Type of Measurement	Unit Conversions in the Customary System
Length	$1 \text{ foot (ft)} = 12 \text{ inches (in.)}$ $1 \text{ yard (yd)} = 3 \text{ feet (ft)}$ $1 \text{ mile (mi)} = 5,280 \text{ feet (ft)}$ $1 \text{ mile (mi)} = 1,760 \text{ yards (yd)}$
Capacity	$1 \text{ cup (c)} = 8 \text{ fluid ounces (fl oz)}$ $1 \text{ pint (pt)} = 2 \text{ cups (c)}$ $1 \text{ quart (qt)} = 2 \text{ pints (pt)}$ $1 \text{ gallon (gal)} = 4 \text{ quarts (qt)}$
Weight	$1 \text{ pound (lb)} = 16 \text{ ounces (oz)}$ $1 \text{ ton} = 2,000 \text{ pounds (lb)}$

6.11.2 GUIDED INSTRUCTION: CONVERTING UNITS OF MEASURE USING RATIO RELATIONSHIPS

Component Overview: Students name different customary units of measurement for length, weight, volume, and time and find their unit conversions. Guide students through this component to explore how to convert units of measurement using the methods learned throughout this unit such as double number lines and tables.

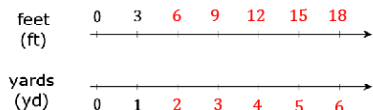
DIFFERENTIATE INSTRUCTION

Consider activating student interest by prompting one of the following:

- What types of units of measure do we think Jared (from the 6.11.1 Warm-Up) might use on a daily basis?
- Brainstorm some units of measure that your family members might use on a daily basis in their jobs.
- What is your dream job? What units of measure are used most often in that environment?

Each conversion rate represents a ratio. Double number line diagrams can be used to convert units.

- Use the double number line diagram shown to find how many feet (ft) are in 4 yards (yd).



There are **12** feet in 4 yards.

Tables can also be used to convert units of measure.

- The average weight of an African forest elephant is 6,000 pounds. Use the unit conversion and the table to determine how many tons the average African forest elephant weighs.

Pounds	Tons
2,000	1
6,000	3

Notice that each conversion rate is a unit rate. Unit rates can be used as factors to convert from one unit to another.

For example, the factor $\frac{4 \text{ quarts}}{1 \text{ gallon}}$ is equivalent to 1 because the numerator and denominator are equivalent values. Multiplying by factors equivalent to 1 will result in equivalent values.

TEACHER MOVES

Think-Pair-Share

Consider implementing a Think-Pair-Share before students use the representations to convert the units. Pose the following questions to students to activate prior knowledge and allow students to verbalize their thinking, which provide an auditory entry point.

- How can you use the double number line to determine how many feet are in 4 yards?
- How can you use the equivalent ratio table to determine how many tons the average African forest elephant weighs?

6.11.2 GUIDED INSTRUCTION: CONVERTING UNITS OF MEASURE USING RATIO RELATIONSHIPS (CONTINUED)

QUESTIONING

Challenge students who may finish early to convert units using the metric system or units of time. Consider asking students to determine the number of minutes in a week or a year (like in the musical Rent – 525,600). This will provide additional practice with more difficult conversion rates.

Kaluba determined how many quarts are in 2 gallons of milk using this strategy. His work is shown.

$$\frac{2 \text{ gal}}{1} \times \frac{4 \text{ qt}}{1 \text{ gal}} = 8 \text{ qt}$$

Kaluba explained, "In my calculation, the unit 'gallons' is in the numerator and the denominator. This means I can eliminate this unit because it becomes 1, similar to when equivalent values are in the numerator and denominator. The result of the conversion is then the number of quarts."

5. Complete the calculation to find how many fluid ounces (fl oz) are in 3 pints (pt).

$$\frac{3 \text{ pt}}{1} \times \frac{2 \text{ cups}}{1 \text{ pt}} \times \frac{8 \text{ fl oz}}{1 \text{ cup}} = 48 \text{ fl oz}$$



The units of measure used in the United States are **customary units**.



The units of measure used in most of the world are **metric units**. Like the decimal system, the metric system uses the base 10.

The tables below show additional unit conversion rates for other units of measure.

Type of Measurement	Unit Conversions in the Metric System
Length	1 meter (m) = 100 centimeters (cm) 1 meter (m) = 1000 millimeters (mm) 1 kilometer (km) = 1,000 meters (m)
Capacity	1 liter (L) = 1000 milliliters (mL)
Weight or Mass	1 gram (g) = 1000 milligrams (mg) 1 kilogram (kg) = 1,000 grams (g)
Time Conversions	
1 minute (min) = 60 seconds (s) 1 hour (h) = 60 minutes (min) 1 day = 24 hours (h) 1 year = 365 days 1 year = 52 weeks 1 week = 7 days	



6.11.3 Guided Practice: Converting Units of Measure Using Unit Rates

- Use the conversion rates to find each value. Show your work.
 - $2.62 \text{ km} = 2620 \text{ m}$
 - $9 \text{ in} = \frac{1}{4} \text{ yd}$
 - $1 \text{ week} = 168 \text{ hours}$
- Adrian weighed 7 pounds (lb) and 4 ounces (oz) when he was born. How many ounces did Adrian weigh when he was born?
116 oz
- Ibuprofen is a medicine sometimes used for pain relief, such as for a headache. For adults, it is not safe to take more than 3200 milligrams (mg) of ibuprofen each day. How many grams of ibuprofen is safe for adults to take in one day?
3.2 grams

6.11.3 GUIDED PRACTICE: CONVERTING UNITS OF MEASURE USING UNIT RATES

Component Overview: Students work with a partner to convert units of measurement using unit rates.

TEACHER MOVES

Think-Pair-Share for MA.K12.MTR.1.1 Practice

Consider structuring this component with the Think-Pair-Share routine.

- First, have students answer the questions individually, emphasizing their process rather than their product.
- Then, have students discuss their answers. Tell students to justify their responses and critique each other's reasoning.
- Discussing the questions will help students synthesize their understanding and build student confidence.

6.11.4 COLLABORATION: ERROR ANALYSIS

Component Overview: Students work with a partner or small group to analyze errors in a student's work converting units of measure.

DIFFERENTIATE INSTRUCTION

If students have not mastered the multiplication and division facts, let them use a calculator to determine the ratios, rates, and percentages. The calculator will allow students to focus on mastering the learning goals of the lesson.

TEACHER MOVES

Circulate as students work with their partners to make informal observations and listen as students engage in discourse. Students should be using math vocabulary to find and explain the errors in Valerie's work, such as conversion, unit rate, and ratio.

ENRICHMENT

Students may struggle to explain Valerie's mistakes in writing or struggle to explain a strategy to fix them. Ask probing questions to guide student thinking or convert these questions into sentence stems for students to guide their responses.

- What did Valerie do wrong? How did you fix that?
- What if you complete the question on your own and compare your work to Valerie's? Find the differences.
- How can you use units to help convert measurements correctly?



6.11.4 Collaboration: Error Analysis

1. Valerie makes sure she drinks 8 cups (c) of water each day. She wondered how many gallons of water this is, so she used conversion rates to find out. Her work is shown.

$$\frac{80 \text{ c}}{1} \times \frac{2}{1} \times \frac{1}{2} \times \frac{1}{4} = \frac{160}{8} = 20 \text{ gal}$$

Valerie knew her answer didn't make sense because there is no way she drinks 20 gallons of water each day. She is not sure where she went wrong.

- a. Discuss Valerie's work with your partner. Then, work together to find the correct number of gallons she drinks each day.

Valerie's first mistake was inputting 80 cups instead of her daily intake of 8 cups. Also, Valerie's first conversion factor, $\frac{2}{1}$, should be $\frac{1}{2}$ because there are 2 cups in 1 pint.

The correct answer is:

$$\frac{8 \cancel{\text{c}}}{1} \times \frac{1 \cancel{\text{pt}}}{2 \cancel{\text{c}}} \times \frac{1 \cancel{\text{qt}}}{2 \cancel{\text{pt}}} \times \frac{1 \text{ gal}}{4 \cancel{\text{qt}}} = \frac{8 \text{ gal}}{16} = 0.5 \text{ gal}$$

- b. Help Valerie understand and learn from her mistakes. Write a brief note to Valerie explaining what her mistakes were and how she can learn from them. Include strategies that may be helpful.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

Hello Valerie. I see that you do know your conversions! The one mistake you made in this problem is in the first conversion factor, $\frac{2}{1}$, which should be $\frac{1}{2}$ because there are 2 c in 1 pt. By writing $\frac{2}{1}$, you will not be able to eliminate the cups unit. One strategy that may be helpful is to include the units when writing the conversion factors. This will help you see that you are eliminating the correct units and using the correct conversion factors.

- c. Read your partner's note to Valerie and offer suggestions on ways to improve their note.
Answers will vary.



6.11.5 Your Turn!

- Use the conversion rates to find each value. Show your work.
 - 11,200 milliliters = **11.2** liters
 - 3.5 gallons = **56** cups
 - 2 days = **2,880** minutes
- Malik plays football for Jones High School in Orlando, Florida. Last Friday, he ran 94 yards for a touchdown on a kickoff return. How many feet did Malik run on the kickoff return?

$$\frac{94 \text{ yd}}{1} \times \frac{3 \text{ ft}}{1 \text{ yd}} = \frac{282 \text{ ft}}{1} = 282 \text{ ft}$$
- About 727 pints of milk are distributed each day from the cafeteria at Fort King Middle School in Marion County, Florida. How many gallons of milk are distributed at Fort King Middle School each day?

$$\frac{727 \text{ pt}}{1} \times \frac{1 \text{ qt}}{2 \text{ pt}} \times \frac{1 \text{ gal}}{4 \text{ qt}} = \frac{727 \text{ gal}}{8} = 90.875 \text{ gal}$$

6.11.5 YOUR TURN!

Component Overview: Students work independently to practice converting units of measure using unit rates.

ENRICHMENT

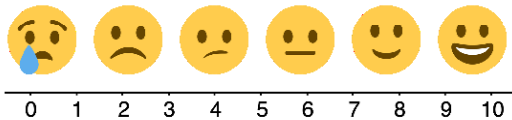
To encourage mathematical reasoning as students complete this component, ask questions like:

- How can you determine the reasonableness of your answer using unit rates?
- How can you estimate what your answer should be before converting? Why would this be helpful?
- Will Malik have run more in feet or yards? How can you use this to assess the reasonableness of your response?



6.11.6 Wrap-Up: Reflection

- Use the emoji scale to indicate your level of confidence with using ratios to convert units within the same measurement system.



Answers will vary.

- Explain why you indicated the confidence level above.
 Answers will vary.

6.11.6 WRAP-UP: REFLECTION

Component Overview: Consider using all or parts of this exercise as a formative assessment to gather data on student progress from this lesson. This Wrap-Up assesses students' level of confidence with using ratios to convert units within the same measurement system.

DIFFERENTIATE INSTRUCTION

Encourage students to get creative and represent their learning or progression from the lesson using video app platforms, drawings, or other methods and submit as homework.